Home Equity Loan Default Prediction

A Credit-Risk Model Recommendation for Underwriting Decision Support

Prepared for: Consumer Credit Department, Credit Risk & Model Risk Governance

Purpose of This Report

This report translates a predictive-modeling analysis of the bank's historical Home Equity Line of Credit (HELOC) portfolio into a decision-support recommendation. It is written for credit-risk, underwriting, and model-risk stakeholders rather than a technical audience — model mechanics are summarized only to the extent needed to assess business fit, governance implications, and deployment risk.

1. Business Context and Problem

A substantial share of retail bank profitability comes from interest income on home-secured lending. Defaults on this book (non-performing assets) directly erode that profitability, and the current underwriting process — a manual file review — is slow, inconsistent across underwriters, and exposed to human error and unintentional bias. Regulatory expectations under the Equal Credit Opportunity Act (ECOA) require that any credit-scoring model be empirically derived, statistically sound, and capable of supporting a specific, defensible reason for any adverse action (a decline).

This analysis was conducted on 5,960 historical HELOC applicants (the hmeq portfolio extract), of whom 1,189 (19.95%) ultimately defaulted or became severely delinquent. The objective was to build a classification model that identifies likely defaulters at the point of underwriting and to surface the applicant characteristics the bank should weight most heavily in that decision.

2. What the Data Shows: Key Risk Drivers

The analysis converged — across three independent methods (a gradient-boosted model's importance ranking, a model-agnostic permutation test, and a fully transparent logistic-regression coefficient view) — on a consistent, credit-intuitive set of drivers. The table below should be read as "what the model relies on to predict default," not as proof of a causal mechanism; these are observed statistical associations in historical data.

Risk Driver	Direction	What the Data Shows
DEBTINC missing	Strongly ↑ risk	62.0% default rate when debt-to-income could not be established, vs. 8.6% when reported — the single strongest signal found in this analysis.
DELINQ	↑ risk	More delinquent credit lines associates with materially higher default rates; the strongest linear correlate of BAD (r=0.354).
DEROG	↑ risk	More major derogatory reports associates with higher default risk (r=0.276).
DEBTINC (reported value)	↑ risk	Among applicants who report it, a higher debt-to-income ratio associates with higher default risk (r=0.200).
CLAGE	↓ risk	A longer credit history (older oldest credit line) associates with lower default risk (r=-0.170).
JOB category	Mixed	Self-employed and Sales occupations show elevated default rates; Professional/Executive and Office show lower rates.

The standout finding, worth escalating to Credit Risk and Data Operations independent of any model decision, is that applicants for whom debt-to-income could not be established defaulted at more than seven times the rate of applicants who reported it (62.0% vs. 8.6%). This may reflect incomplete documentation, self-reporting avoidance,

or an underwriting process that historically did not chase down DTI for already-marginal files. The data cannot distinguish between these explanations, and we recommend a targeted operational review rather than assuming a single cause.

3. Modeling Approach and Results

Four model families were required by the project scope and evaluated on an identical, leakage-safe 70/30 train-test split (all imputation, scaling, and outlier-treatment statistics were derived from training data only): Logistic Regression, Decision Tree, Random Forest, and XGBoost (gradient boosting). Because a missed defaulter (a "False Negative" — the model predicts repayment but the applicant defaults) is far more costly to the bank than a wrongly-flagged good applicant (a "False Positive" — foregone interest income, recoverable via manual override), every model was evaluated primarily on Recall — the share of actual defaulters correctly identified — rather than on overall Accuracy, which is misleading on a book where 80% of applicants repay by default.

Model	Recall	Precision	F1	ROC-AUC	CV Recall
Logistic Regression (Tuned)	0.742	0.545	0.629	0.868	0.779
Decision Tree (Tuned)	0.804	0.554	0.656	0.886	0.828
Random Forest (Tuned)	0.770	0.668	0.715	0.933	0.801
XGBoost (Tuned) — Recommended	0.798	0.764	0.781	0.950	0.835

The recommended model — XGBoost, tuned via cross-validated hyperparameter search toward Recall, with preprocessing refit independently within each cross-validation fold to avoid data leakage — identifies approximately 80% of actual defaulters in the held-out test sample, while keeping over 76% of its "predicted default" flags correct. This was not selected purely for having the top ROC-AUC (a measure of overall ranking ability); Random Forest scored close behind on that metric (0.933 vs. 0.950). XGBoost was selected because it led on Recall, Precision, and F1 simultaneously, and because its risk-driver rankings could be independently cross-checked against the fully transparent Logistic Regression model — giving both strong performance and a credible interpretability path. Its output should be read as a risk-ranking score, not a literal probability of default: the model discriminates well between likely and unlikely defaulters, but a direct check of its predicted probabilities against observed outcomes found they are not well calibrated in the middle of the risk range (see Section 5).

4. Recommended Deployment Approach

- Use the model as underwriter decision support (a risk score and flag), not as a fully automated approve/decline system. Human review remains the final step, consistent with regulatory expectations and with the residual uncertainty inherent in any statistical model.
- Set the operating decision threshold in the 0.30–0.40 probability range rather than the conventional 0.50, to reflect the bank’s stated priority on catching defaulters — subject to formal validation by Credit Risk against the bank’s actual loss-given-default and review-cost figures. This range was selected using cross-validated probabilities on the training data and confirmed once on the untouched test set (see Section 3).
- Pair every model-driven decline with a plain-language explanation grounded in the top risk drivers above (e.g., elevated derogatory-report count, undocumented debt-to-income), to support a defensible adverse-action rationale — pending formal compliance sign-off that the explanation satisfies ECOA’s specific-reason requirement.
- Run the model in shadow mode alongside current underwriting for an initial evaluation period before any live decisioning use, to validate real-world performance before committing to production reliance.

5. Model Risk, Limitations, and Required Follow-Up

This section is deliberately conservative — a credible model-risk assessment states limitations plainly rather than only where convenient.

- Selection bias: the model was trained only on applicants who were historically approved. Its behavior on the population that would have been declined under the prior process has not been observed or validated.
- No demographic data: hmeq.csv contains no age, gender, race, or other protected-characteristic fields. This report cannot and does not claim the model is free of disparate impact — a full fair-lending review requires data outside the scope of this analysis, and REASON/JOB (the only categorical fields present) can act as partial proxies for factors this dataset cannot observe directly.
- Calibration, not just discrimination: the model ranks applicants by risk well (ROC-AUC 0.950 on the test set), but a direct calibration check (Brier score and a 10-bin reliability curve) found its raw probability outputs over-state risk in the middle of the range — for example, a mean predicted probability of 0.36 corresponded to an observed default rate of about 0.25. The model's output should be used as a ranking/flagging signal, not quoted to stakeholders as a literal, calibrated probability of default, unless a recalibration step is added.
- No dollar-denominated cost-benefit figures: this dataset does not include interest rates, realized loss amounts, or underwriting review costs. Any financial benefit estimate must be produced by Finance/Credit Risk using their own figures; a calculation framework is provided in the accompanying Final Submission document.
- Model drift: applicant populations and macroeconomic conditions change over time; a periodic retraining and monitoring cadence should be established before reliance on this model becomes routine.
- Preprocessing choices (median/mode imputation, IQR-based outlier capping) are standard and defensible for this analysis but are not the only valid approach, and involve some loss of information at the tails of skewed variables.

6. Recommended Next Steps

- Credit Risk: formally validate and sign off on the operating threshold using actual cost data.
- Compliance / Fair Lending: commission a disparate-impact review with appropriate demographic data before any production deployment.
- Data Operations: investigate the root cause of the DEBTINC collection gap.
- Model Risk Governance: establish a monitoring, retraining, and periodic revalidation cadence consistent with the bank's model risk management policy.